

# Recap

$$H = E[I(x)] = \sum_x p(x) I(x) = - \sum_x p(x) \log p(x)$$

- Entropy

$$H(x, y) = - \sum_{x, y} p(x, y) \cdot \log p(x, y)$$

- Joint Entropy, Conditional Entropy

$$H(Y|X) = \sum_x p(x) H(Y|X=x)$$

an informative feature leads to high  $I(X_i, Y)$

average conditional entropy classification

- Mutual Information

$$I(X_i, Y) = H(Y) - H(Y|X_i)$$

reduction in uncertainty of  $Y$  given we observed feature  $X_i$   
 $X_i$  has a lot of information (weight in cat/dog)

- Cross Entropy

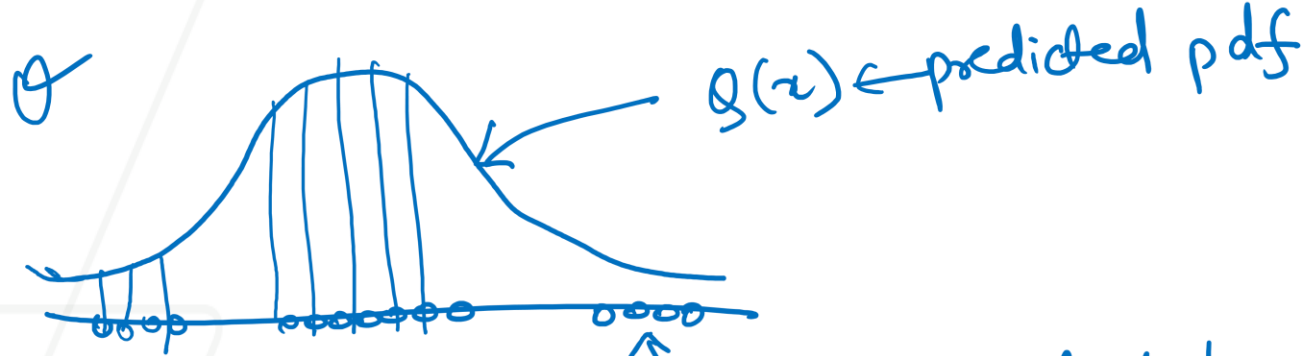
low inefficiency  
we are encoding samples from  $P(x)$  by actually following  $Q(x)$

$$H(Y|X_i) \downarrow \quad H(Y) - H(Y|X_i) \uparrow$$

# Optimization

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$$E[a+b+c] = E[a] + E[b] + E[c]$$

$$E[g(x)] = \sum_x g(x) \cdot p(x)$$

$$L(\theta|x) = \prod_{i=1}^n f(x^{(i)}|\theta)$$

maximize

$$\log L(\theta|x) = \sum_{i=1}^n \log f(x^{(i)}|\theta)$$

minimize

$$l(\theta|x) = - \sum_{i=1}^n \log f(x^{(i)}|\theta)$$

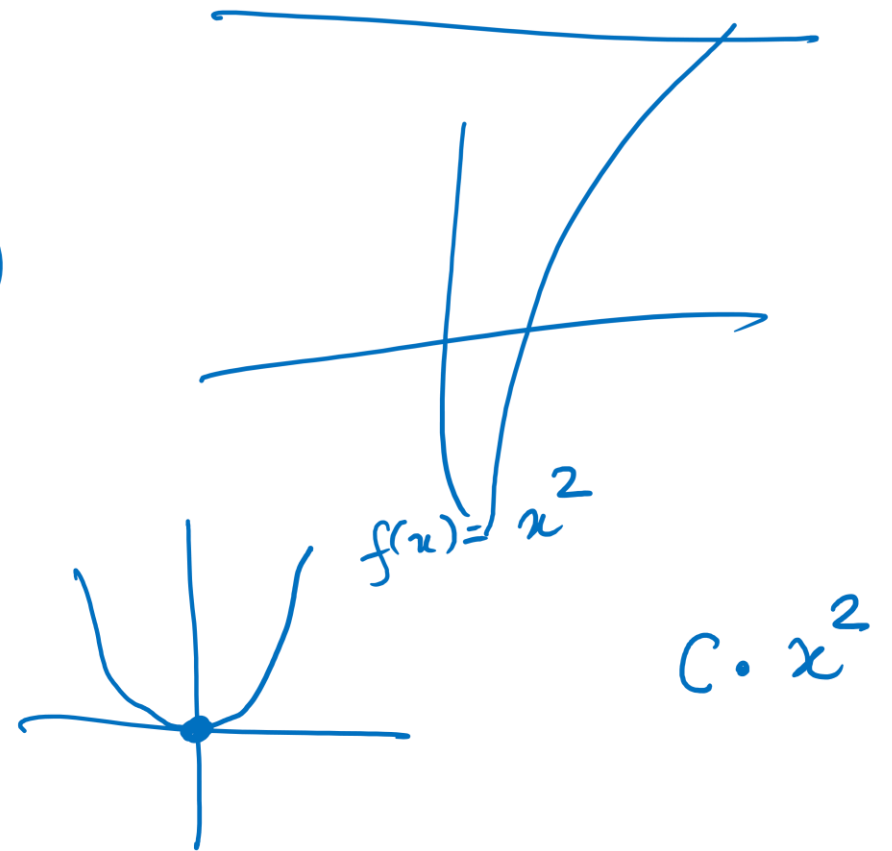
$$= - \frac{1}{n} \sum_{i=1}^n \log f(x^{(i)}|\theta)$$

$$= - E[\log Q(x)]$$

$$= - E[\log P(x) - \log P(x) + \log Q(x)]$$

$$= - E[\log P(x)] - E\left[\log \frac{Q(x)}{P(x)}\right]$$

$$H(P) = - \sum_x p(x) \log p(x) = - \sum_x p(x) \log \frac{Q(x)}{P(x)}$$



①

$$\min l(\theta|x) = \underbrace{-\sum_x p(x) \log p(x)}_{H[p]} - \underbrace{\sum_x p(x) \log \frac{q(x)}{p(x)}}_{KL[p][q]}$$

likelihood minimization = Min.  $H[p] + KL[p][q]$

Minimizing CE is the same as minimizing -ve log likelihood.

# Cross Entropy

**Cross Entropy:** The expected number of bits when a wrong distribution  $Q$  is assumed while the data actually follows a distribution  $P$

$$H(p, q) = - \sum_{x \in \mathcal{X}} p(x) \log q(x) = H(P) + KL[P][Q]$$

*Handwritten annotations:*

- $H(p, q)$  is circled and labeled "CE" (Cross Entropy).
- The entire expression is labeled "loss function".
- The term  $- \sum_{x \in \mathcal{X}} p(x) \log q(x)$  is labeled "objective function".
- The term  $KL[P][Q]$  is labeled "non negative".
- A bracket under the sum term is labeled  $\sum_x p(x) \log \frac{1}{1/q(x)}$ .

This is because:

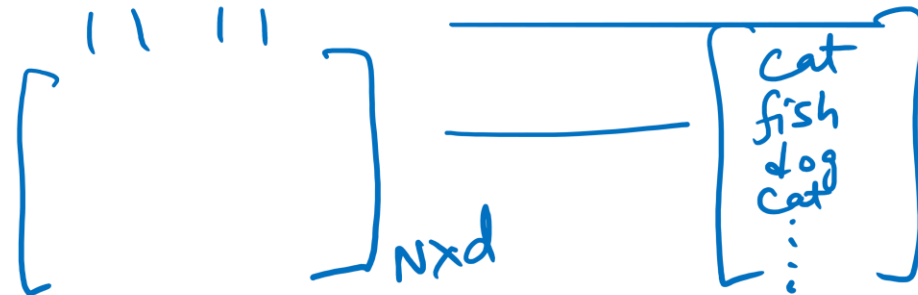
$$H(p, q) = \mathbb{E}_p[l_i] = \mathbb{E}_p \left[ \log \frac{1}{q(x_i)} \right]$$

$$H(p, q) = \sum_{x_i} p(x_i) \log \frac{1}{q(x_i)}$$

$$H(p, q) = - \sum_x p(x) \log q(x).$$

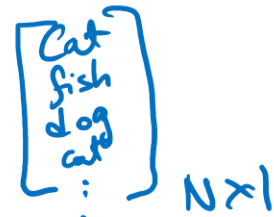
# Labeling target values: Label encoding (ordinal) and One-hot encoding

• Input Matrix (X):



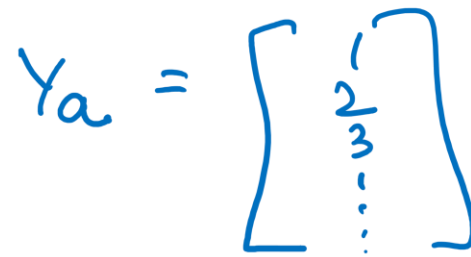
Multiclass classification

• Target Labels (Y):

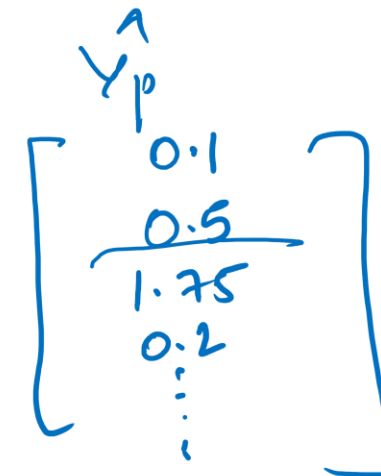


• Label Encoding (Ordinal):

cat  $\rightarrow$  1, fish  $\rightarrow$  2, dog  $\rightarrow$  3



model

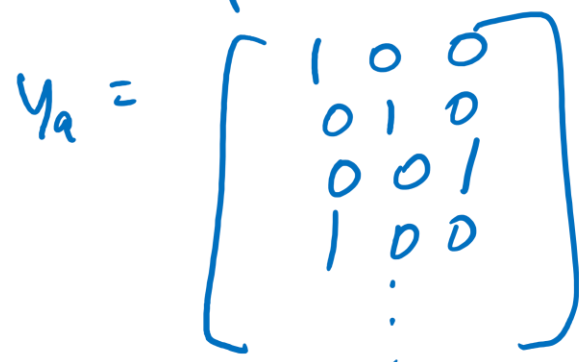


tshirt sizes

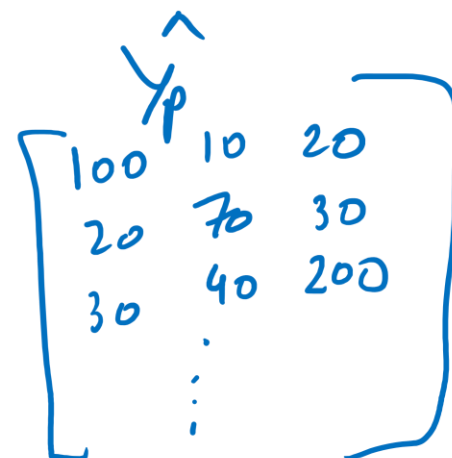
• One-Hot Encoding:

Cat  $\rightarrow$  [1 0 0], fish  $\rightarrow$  [0 1 0], dog  $\rightarrow$  [0 0 1]

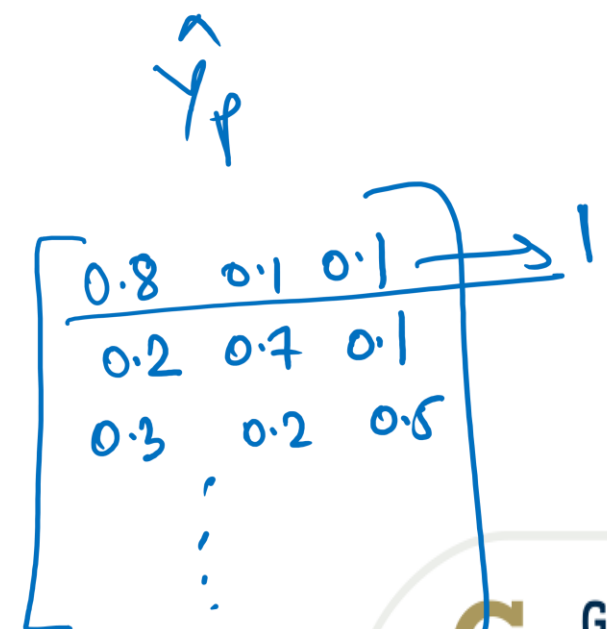
sums up to 1  
pm f



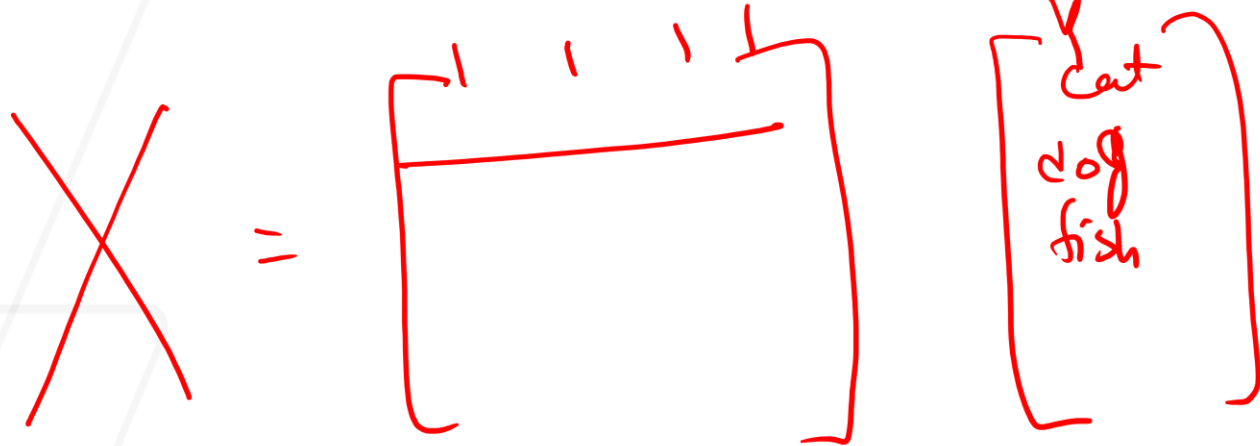
model



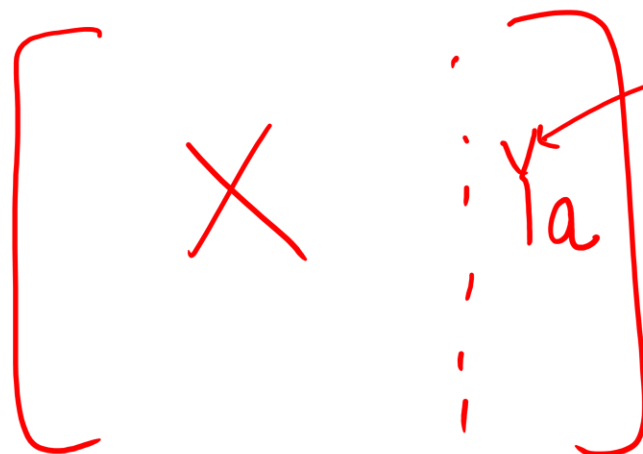
Softmax







Training data



actual labels

Model

use first set of parameters that it assumes



predicted target values

How  $Y_a$  and  $\hat{Y}_p$  vary

as close as possible

Trained Model  $\rightarrow$  a new datapoint  $\rightarrow \hat{Y}_p$

$f(x)$   
Objective

maximize or minimize



# Label encoding (One-hot encoding) Loss Computation

Minimize Objective Function:

$$\min \sum_{i=1}^n \| y_a^{(i)} - \hat{y}_p^{(i)T} \|_2^2$$

MSE

1st data point

$$y_a = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$
$$\hat{y}_p = \begin{bmatrix} 0.8 & 0.1 & 0.1 \end{bmatrix}$$

$$(1 - 0.8)^2 + (0 - 0.1)^2 + (0 - 0.1)^2 = (0.2)^2$$

In a perfect world:  $\hat{y}_p = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$

$$\rightarrow 0 + 0 + 0 + \dots + n = 0$$

Maximize Objective Function:

$$\max \sum_{i=1}^n y_a^{(i)} \cdot \hat{y}_p^{(i)T}$$

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$$\min - \sum_{i=1}^n y_a^{(i)} \cdot \hat{y}_p^{(i)T}$$

$$0.8 \times 1 + 0 \times 0.1 + 0 \times 0.1 = 0.8$$

In a perfect world = 1

$$1 + 1 + 1 + \dots + n = n$$

Interpretation:

- Minimum squared Error works but isn't ideal for classification.
- Minimize negative sum of dot product is also good but Cross-Entropy / NLL is better.



# Why Cross Entropy is better than dot product

$$H(p, q) = - \sum_{x \in \mathcal{X}} p(x) \log q(x)$$

$\uparrow$   $y_a^{\{i\}}$   $\uparrow$   $\hat{y}_p^{\{i\}}$

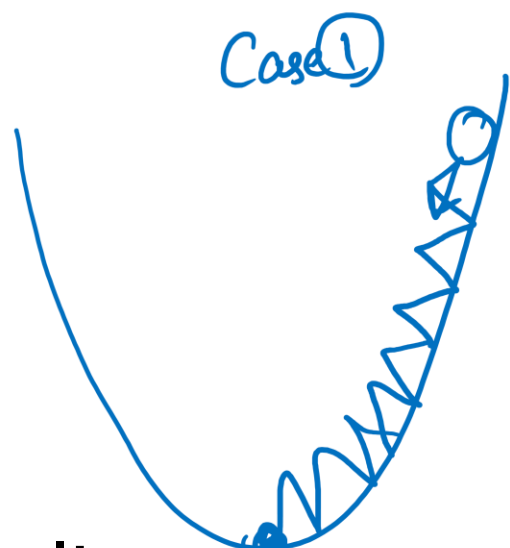
$= \text{CE} = - \sum_{i=1}^n y_a^{\{i\}} \cdot \log \hat{y}_p^{\{i\}T} \quad \text{Case ②}$

$\text{min} \quad - \sum_{i=1}^n y_a^{\{i\}} \cdot \hat{y}_p^{\{i\}T} \quad \text{Case ①}$

$\text{-ve log likelihood}$



Log really really penalizes our objective function



Convergence is quick

Logarithmic penalty encourages big update leading to faster convergence

# Kullback-Leibler Divergence

Another useful information theoretic quantity measures the difference between two distributions.

$$\begin{aligned}\mathbf{KL}[P(S)||Q(S)] &= \sum_s P(s) \log \frac{P(s)}{Q(s)} \\ &= \underbrace{\sum_s P(s) \log \frac{1}{Q(s)}}_{\text{cross entropy}} - \mathbf{H}[P] = H(P, Q) - H(P)\end{aligned}$$

Excess cost in bits paid by encoding according to  $Q$  instead of  $P$ .

KL Divergence is  
a **KIND OF**  
distance  
measurement

$$-\mathbf{KL}[P||Q] = \sum_s P(s) \log \frac{Q(s)}{P(s)}$$

log function is  
concave or  
convex?

$$\begin{aligned}\sum_s P(s) \log \frac{Q(s)}{P(s)} &\leq \log \sum_s P(s) \frac{Q(s)}{P(s)} \\ &= \log \sum_s Q(s) = \log 1 = 0\end{aligned}$$

By Jensen Inequality

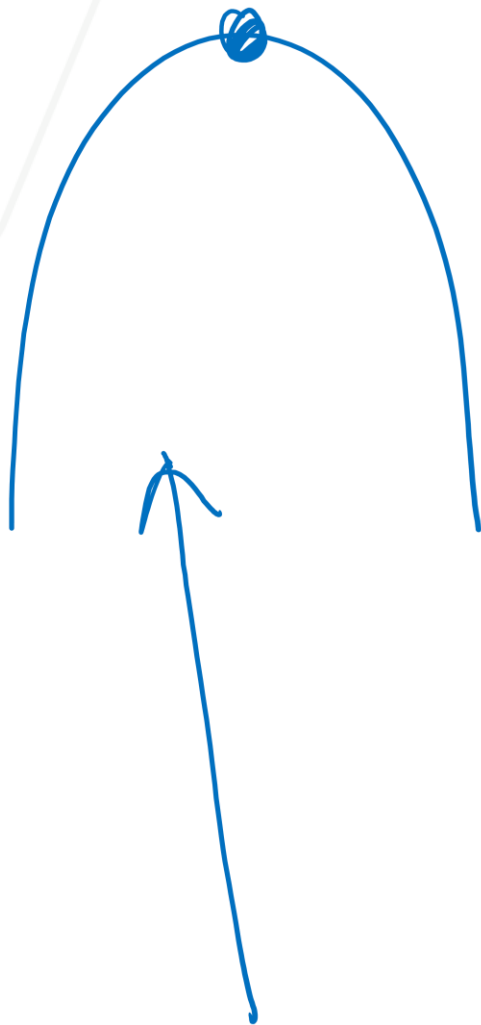
So  $\mathbf{KL}[P||Q] \geq 0$ . Equality iff  $P = Q$

When  $P = Q$ ,  $KL[P||Q] = 0$

# Concave Function: Jensen Inequality

$$\mathbb{E}[f(X)] \leq f(\mathbb{E}[X])$$

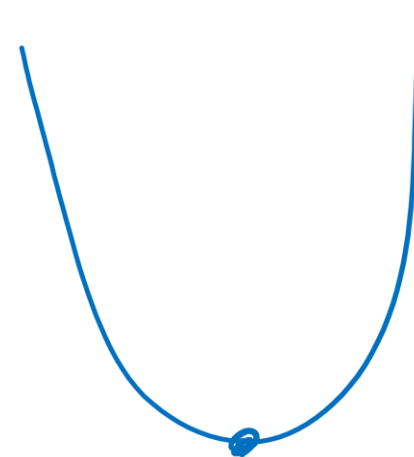
$$f(x) = -x^2$$
$$f''(x) < 0 \rightarrow \text{concave}$$



Strictly concave functions have global maximum

# Convex Function: Jensen Inequality

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$$



$$f(x) = x^2$$

$$x = \begin{bmatrix} 1 & 2 \end{bmatrix}$$

$$p(x) = \begin{bmatrix} 1/2 & 1/2 \end{bmatrix}$$

$$\begin{aligned} \mathbb{E}[f(x)] &= \sum_x f(x) \cdot p(x) = f(x=1)p(x=1) + f(x=2)p(x=2) \\ &= 1 \cdot 1/2 + 4 \cdot 1/2 = 2.5 \end{aligned}$$

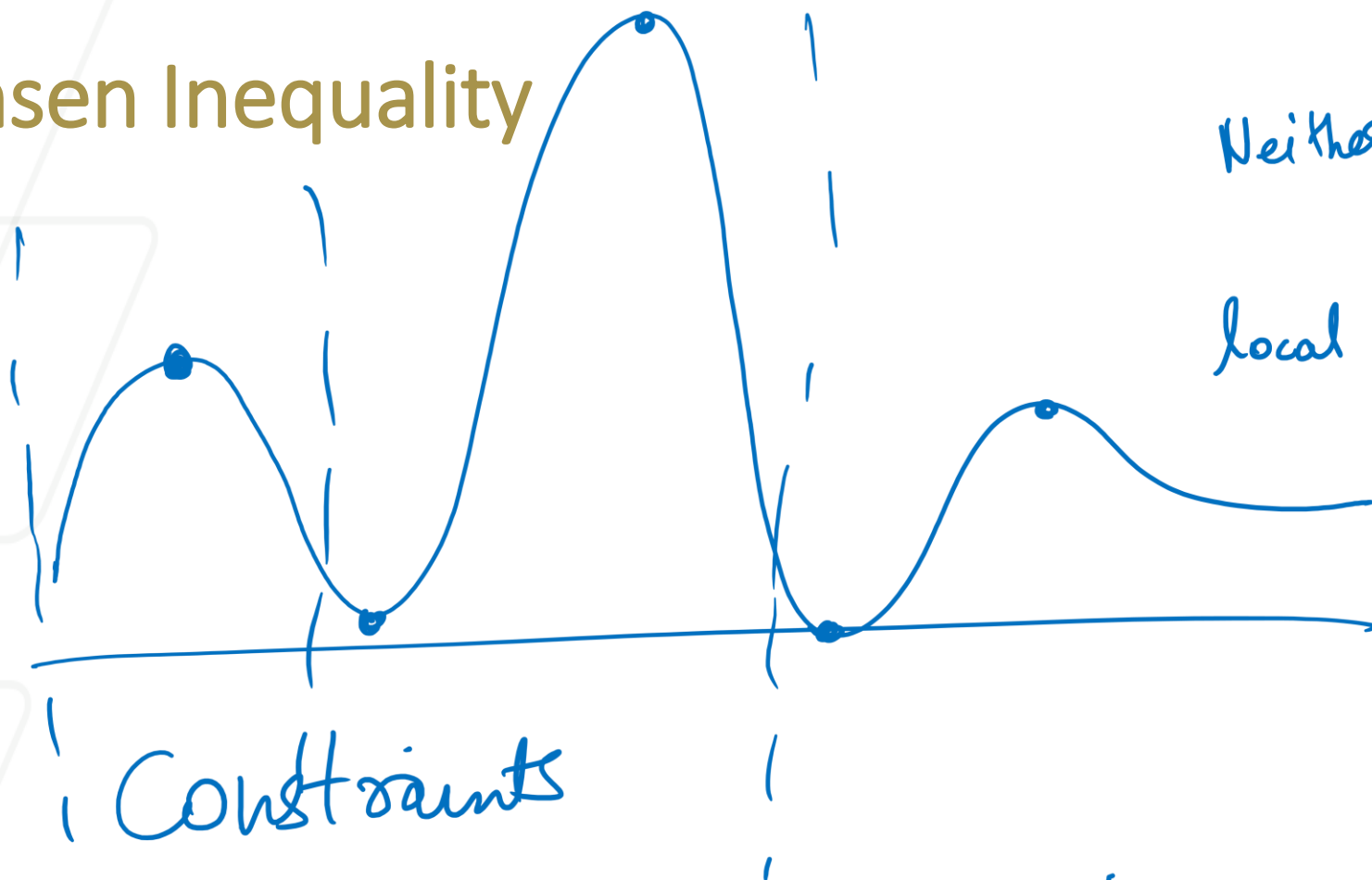
$$f(\mathbb{E}[x]) \quad \mathbb{E}[x] = \sum_x x \cdot p(x) = 1 \times \frac{1}{2} + 2 \times \frac{1}{2} = 1.5$$

Strictly convex functions have global minimum

# Jensen Inequality

Neither concave & convex

local maxima & minima



$$\log(x) =$$
$$f'(x) = 1/x$$
$$f''(x) = -1/x^2$$



Concave.

$$E[f(x)] \leq f(E(x))$$

$$E[\log x] \leq \log(E(x))$$

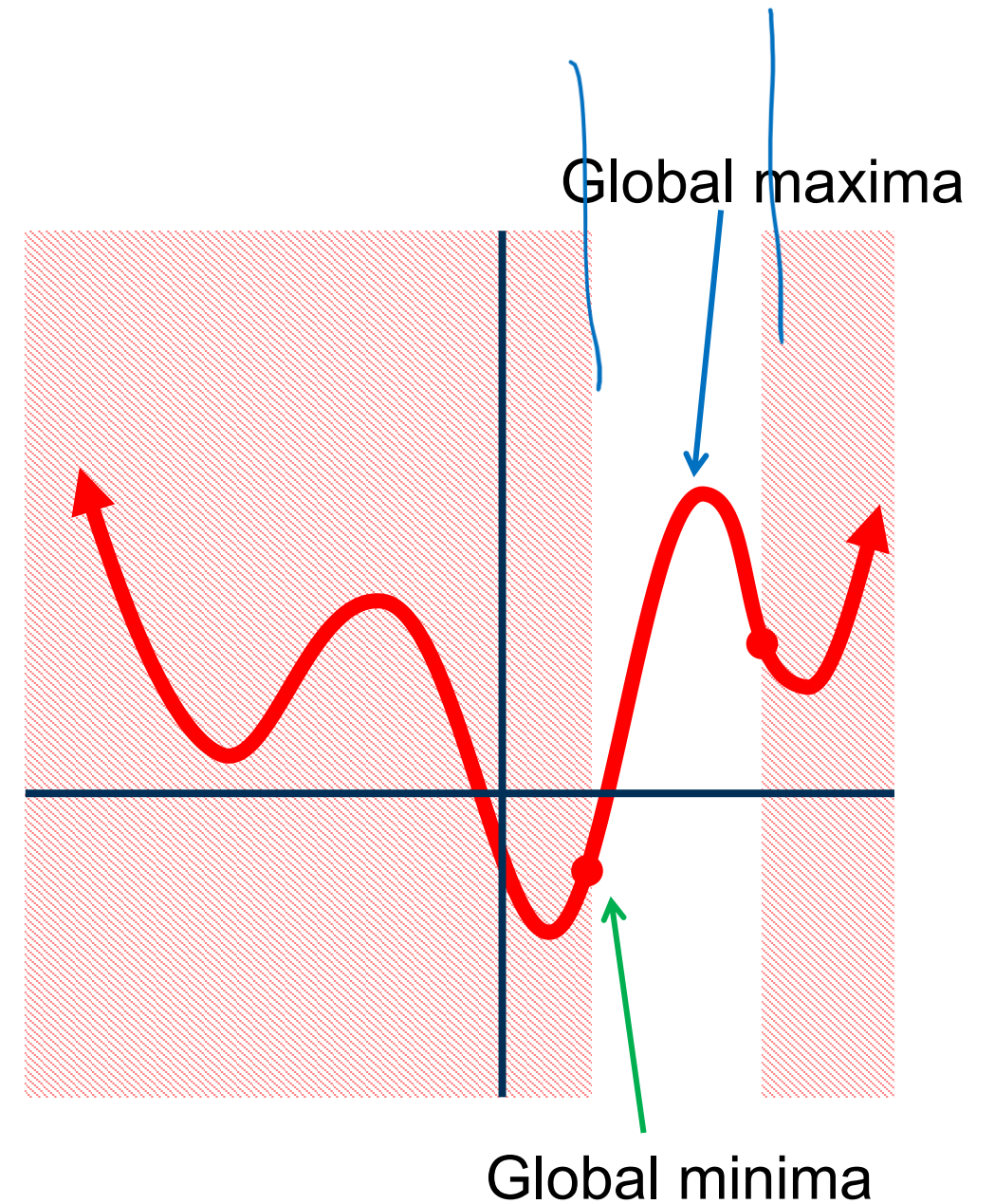
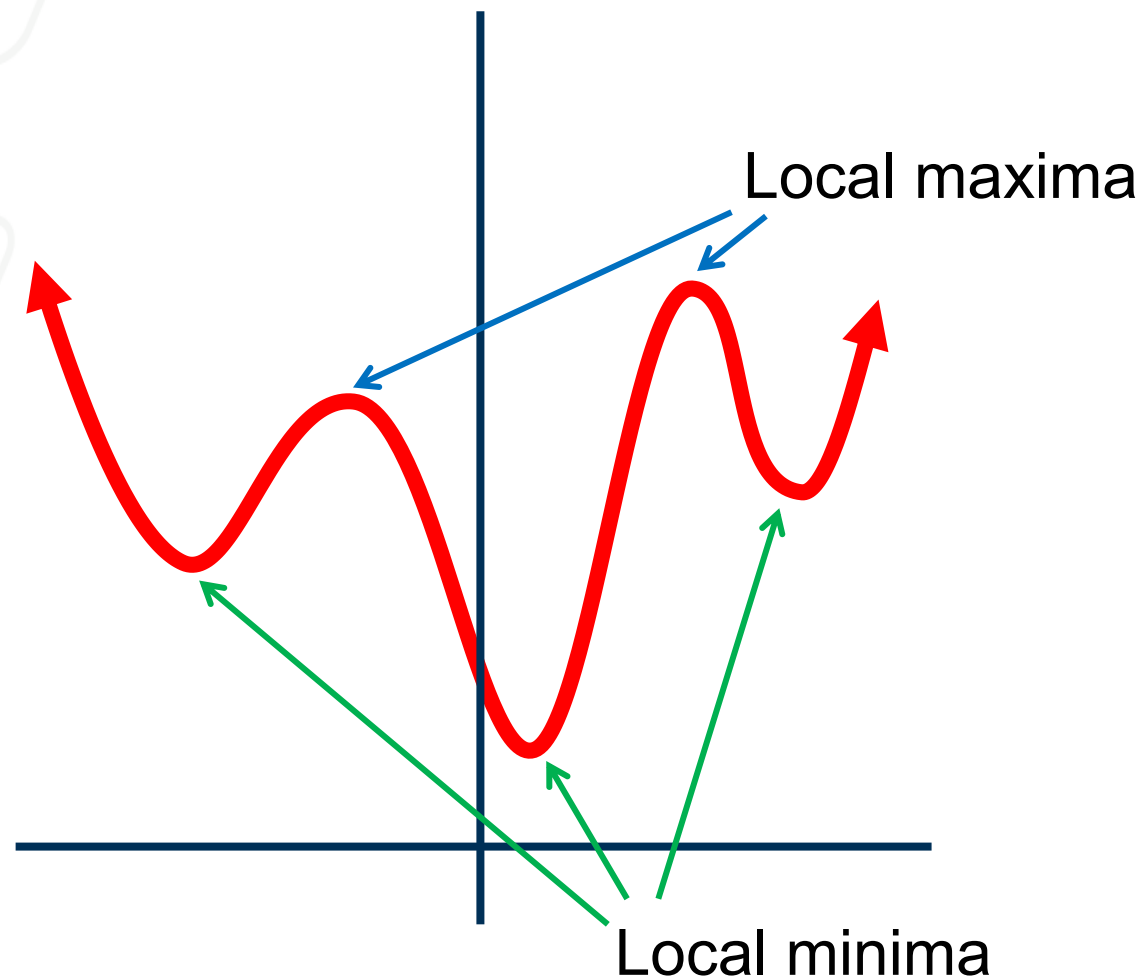
KL Divergence is always non negative

$$\begin{aligned}
 -KL[P][Q] &= \sum_x p(x) \log \frac{q(x)}{p(x)} \\
 &= \sum_x p(x) g(x) \\
 &= E[g(x)] \\
 &= \log(E[q(x)]) \\
 &\geq \log\left(\sum_x p(x) q(x)\right) \\
 &\geq \log\left(\sum_x p(x) \cdot \frac{q(x)}{p(x)}\right) \\
 &\geq \log \sum_x q(x) \leftarrow \text{pdf} \\
 &\geq \log 1 \\
 &\geq 0
 \end{aligned}$$

$-KL[P][Q]$   
 $KL[P][Q] \geq 0$



# Unconstrained and constrained optimization



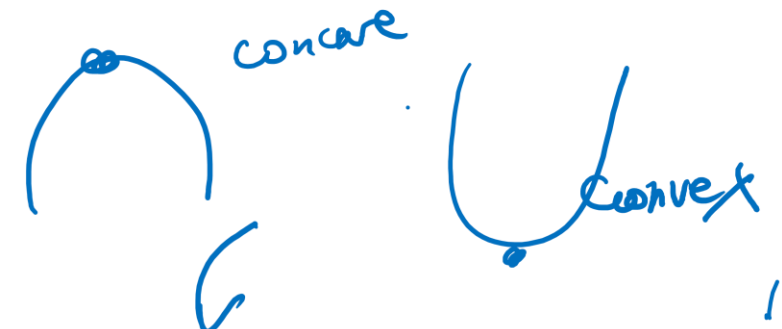
$$f(x,y) = 6x^2 + 5y^2$$

$$\text{Hessian} \begin{bmatrix} \frac{\partial^2 f(x,y)}{\partial x^2} & \frac{\partial^2 f(x,y)}{\partial x \partial y} \\ \frac{\partial^2 f(x,y)}{\partial x \partial y} & \frac{\partial^2 f(x,y)}{\partial y^2} \end{bmatrix}$$

$$\det(\text{Hessian}) < 0 \rightarrow \text{Concave}$$

$$> 0 \rightarrow \text{Convex}$$

# Basic unconstrained optimization problem



- Objective  $f(\mathbf{x})$  the quantity we are trying to optimize (maximize or minimize)
- The variables  $x_1, x_2, \dots, x_m$  which can be represented in vector form as  $\mathbf{x}$   
(Note: the subscript  $m$  indicates a feature NOT a datapoint in our dataset)
- Finding the stationary points where the function is no longer increasing or decreasing

# Equality constrained optimization

$$\min f(x, y) = \frac{6x^2 + 5y^2}{\text{Convex}}$$

→ s.t.  $x + y = 1$

- An equality constrained optimization problem is expressed as:

$$\begin{array}{ll} \min_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}) & \text{objective} \\ \text{subject to } \underline{g(\mathbf{x}, \mathbf{y}) = 0} & \text{s.t.} \end{array}$$

$\underline{g(x, y) = x + y - 1 = 0}$

# Inequality constrained optimization

- An inequality constrained optimization problem is expressed as:

$$\begin{array}{ll} \min_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}) \\ \text{subject to } \underline{g(\mathbf{x}, \mathbf{y}) \leq 0} \end{array}$$

K.K.T. conditions

# Linear, Quadratic, and Nonlinear Programming

## Linear Programming (LP)

**Objective:** Linear function of  $x$ .

**Constraints:** Linear.

**Example:**  $\min 3x_1 + 5x_2 \text{ s.t. } x_1 + 2x_2 \leq 6$

## Quadratic Programming (QP)

**Objective:** Quadratic

**Constraints:** Linear.

**Example:**  $\min x^2 + y^2 + 3x + 5y \text{ s.t. } \underline{x + y \leq 4}$

## Nonlinear Programming (NLP)

**Objective:** Nonlinear.

**Constraints:** Can be nonlinear.

**Example:**  $\min x^2 + y^2 + 3x + 5y \text{ s.t. } x^2 + y \leq 4$



# Recap

- Cross Entropy
- KL Divergence
- Loss Functions
- Jensen Inequality
- Unconstrained and Constrained Optimization
- Linear, Quadratic and Non-Linear Programming

# Equality constrained optimization problem

There is a notion of a Lagrangian expression where we combine both the objective function and our constraints

$$L(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda g(\mathbf{x})$$

$$s.t. \lambda \neq 0$$

# Equality constrained optimization problem

- With multiple equality constraints the Lagrangian is expressed as follows:

$$L(\mathbf{x}, \lambda) = f(\mathbf{x}) - \sum_{i=1}^k \lambda_i g_i(\mathbf{x})$$
$$s. t. \quad \lambda_i \neq 0$$



# Equality constrained optimization problem

- To solve this optimization problem, we only need to satisfy the stationarity condition, which states:

$$\nabla L(\mathbf{x}, \lambda) = 0$$

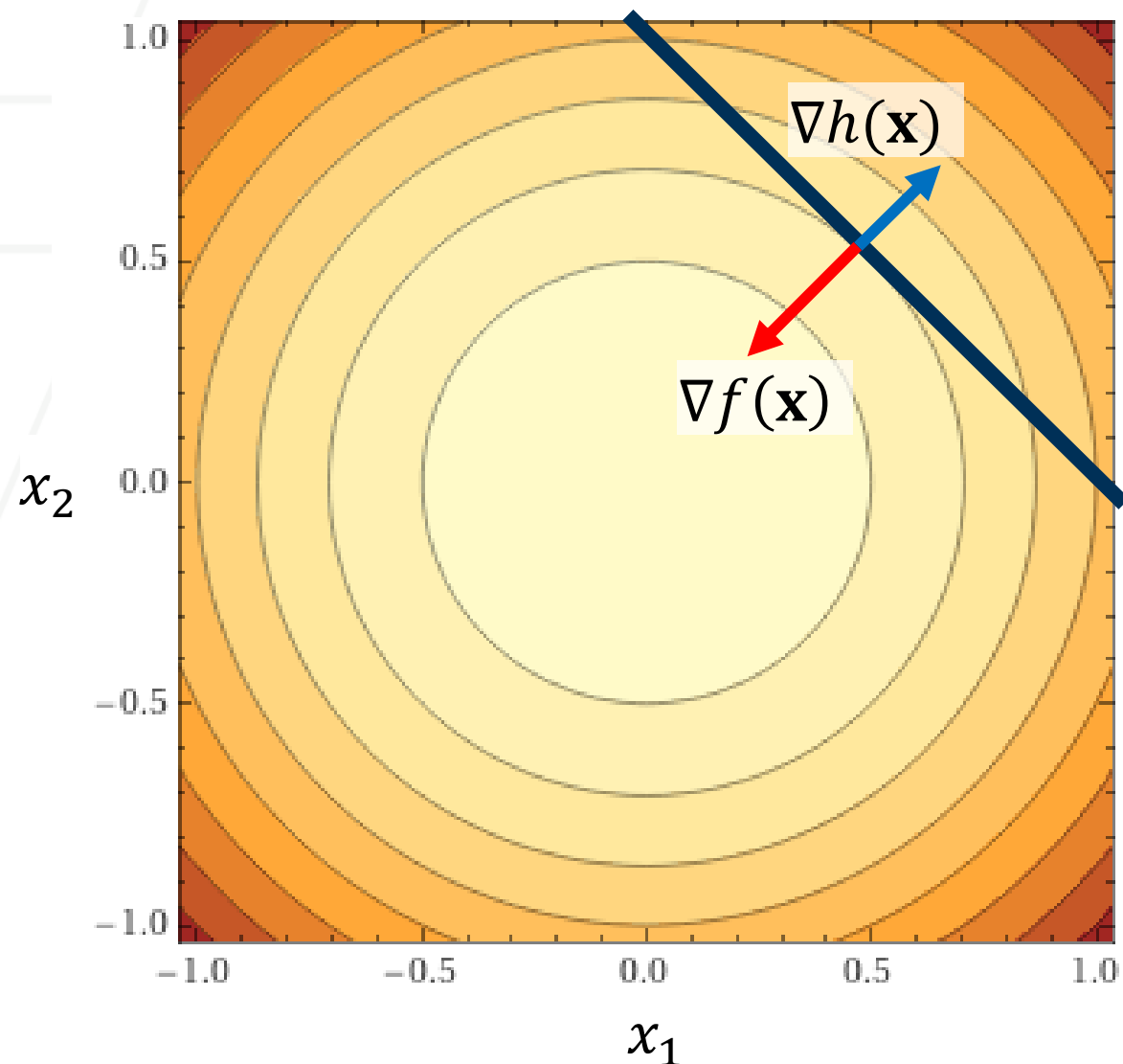
- We also need to satisfy

$$\lambda_i \neq 0$$

- This appeals to the intuition that at the optimal point, the gradient of the equality constraint is proportional to the gradient of the objective function

# Equality constrained optimization: *Example 1*

# Equality constrained optimization: *Example 2*



$$\begin{aligned} \max_{\mathbf{x}} \quad & 1 - x_1^2 - x_2^2 \\ \text{s.t.} \quad & x_1 + x_2 = 1 \end{aligned}$$

# Equality constrained optimization: *Example*

$$\begin{array}{ll}\max_{\mathbf{x}} & 1 - x_1^2 - x_2^2 \\ \text{s.t.} & x_1 + x_2 = 1\end{array}$$

1. Write the Lagrangian for this problem

$$L(\mathbf{x}, \lambda) = 1 - x_1^2 + x_2^2 + \lambda(x_1 + x_2 - 1)$$

2. Calculate the gradient with respect to  $\mathbf{x}$

$$\frac{\partial L}{\partial x_1} = -2x_1 + \lambda = 0 \rightarrow \lambda = 2x_1$$

$$\frac{\partial L}{\partial x_2} = -2x_2 + \lambda = 0 \rightarrow \lambda = 2x_2$$

# Equality constrained optimization: *Example*

3. Solve for  $\mathbf{x}^*$  and  $\lambda$

Using our previous two derivatives and the equality constraint, we obtain:

$$x_1 + x_2 = 1 \rightarrow x_1 + x_1 = 1 \rightarrow x_1^* = x_2^* = \frac{1}{2}$$

From  $\frac{\partial L}{\partial x_1} = 0$  we find the value of  $\lambda$

$$\lambda = 2x_1 \rightarrow \lambda = 1$$

# Inequality constrained optimization problem

- Constraints that limit how small or big variables can be. These are inequality constraints noted as  $g_j(\mathbf{x})$
- An optimization problem is usually expressed as:

$$\begin{aligned} & \min_{\mathbf{x}} f(\mathbf{x}) \\ \text{subject to } & g_j(\mathbf{x}) \leq 0, j = 1, 2, \dots, m \end{aligned}$$

# Inequality constrained optimization problem

- The optimization problem can be rewritten as

$$\begin{aligned} & \min_{\mathbf{x}} f(\mathbf{x}) + \sum_{j=1}^m \lambda_j g_j(\mathbf{x}) \\ & \text{subject to} \quad \lambda \geq 0 \end{aligned}$$

- We need to satisfy the Karush-Kuhn-Tucker (KKT) conditions on  $\mathbf{x}$ ,  $\lambda$  which is **necessary and sufficient** for optimality



# Karush-Kuhn-Tucker Conditions

- Stationarity condition:

$$\min L(\mathbf{x}, \lambda) = f(\mathbf{x}) + \sum_{j=1}^m \lambda_j g_j(\mathbf{x})$$

- Primal feasibility

$$g_j(\mathbf{x}) \leq 0, j = 1, 2, \dots, m$$

- Dual Feasibility

$$\lambda_j \geq 0$$

- Complementary slackness

$$\lambda_j g_j(\mathbf{x}) = 0, j = 1, \dots, m$$

If  $g(\mathbf{x}) < 0 \rightarrow$  **inactive** solution since  $\lambda = 0$

If  $g(\mathbf{x}) = 0 \rightarrow$  **active** solution since  $\lambda$  is at play and  $> 0$ , which means constraint is active



# Karush-Kuhn-Tucker Interpretations

- The complementary slackness condition applies only to inequality constraints
- It states that for a given inequality constraint...  
$$\dots \text{either } g_j(x) = 0 \text{ or } \lambda_j = 0$$
- Whenever the dual variable  $\lambda_j > 0$  and therefore  $g_j(x) = 0$ , we say that the **constraint is active** or that the **constraint is tight at  $\mathbf{x}$**
- The reverse  $\lambda_j = 0$  and therefore  $g_j(x) < 0$  means that the constraint is inactive, meaning that it is not effectively impacting the solution

**Rule:** For each inequality constraint, **either** it is active and has a positive multiplier, or it is inactive and its multiplier is zero.

# Primal vs. Dual Form

# Primal vs Dual Form

## Primal Problem

Original optimization problem

## Dual Problem

Derived problem as **a function of  $\lambda$**  only.

## Weak Duality

- Dual optimum  $\leq$  primal optimum (for minimization problems).
- Always true.

## Strong Duality

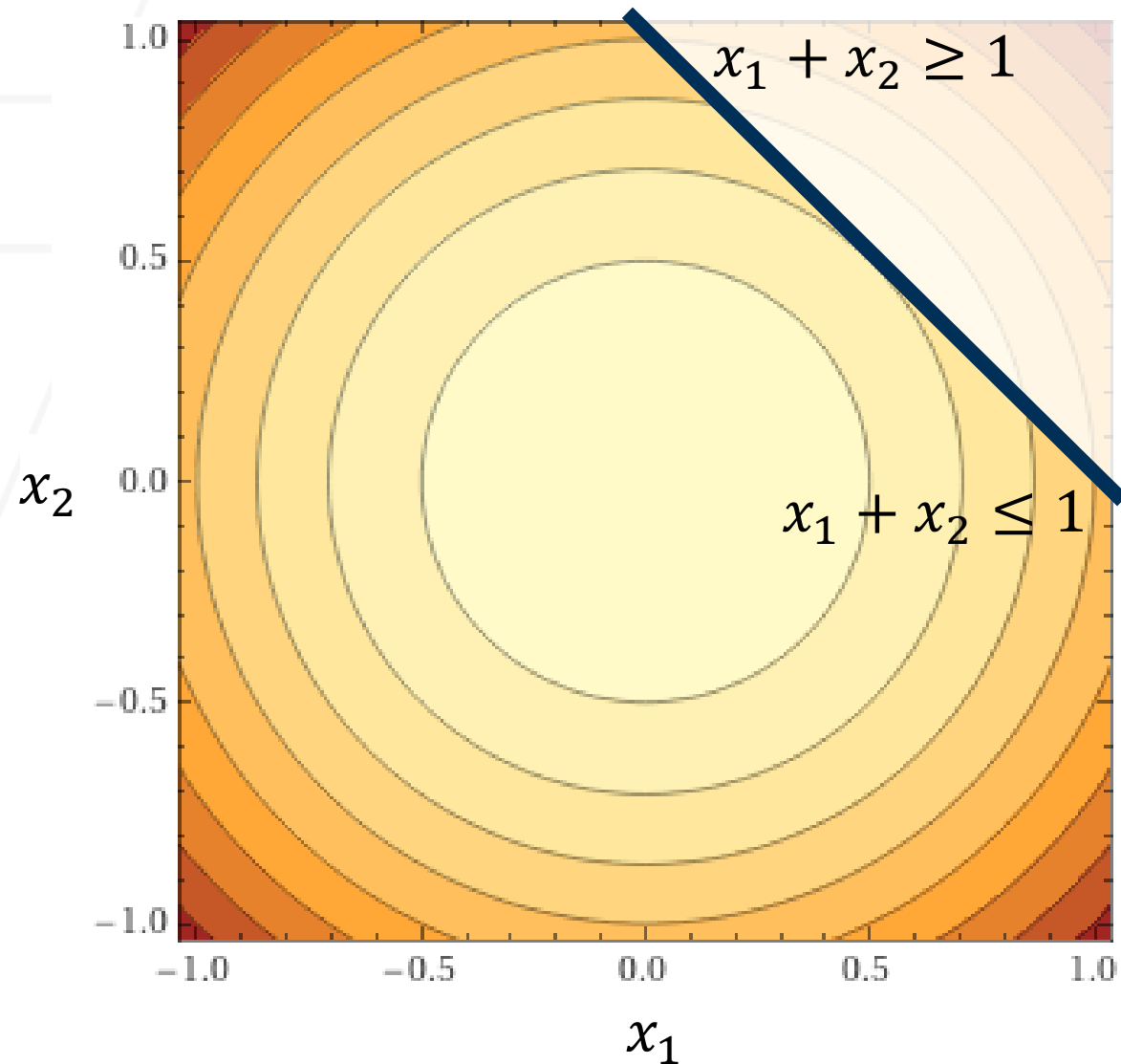
- Dual optimum = primal optimum.
- Holds for **convex problems** with feasible solution.

We like quadratic programming because the objective functions shape allows easier optimization and more chances of minimization due to convex function. So it will always have strong duality.

## Why care?

- Sometimes easier to solve the dual.
- Many ML models (e.g., SVMs) are solved via the dual.

# Inequality constrained optimization: *Example 1*



$$\begin{aligned} \max_{\mathbf{x}} \quad & 1 - x_1^2 - x_2^2 \\ \text{s.t.} \quad & x_1 + x_2 \leq 1 \end{aligned}$$

# Inequality constrained optimization: *Example 1*

1. Write the Lagrangian for this problem

$$L(\mathbf{x}, \lambda) = -1 + x_1^2 + x_2^2 + \lambda(x_1 + x_2 - 1)$$

2. Calculate the gradient with respect to  $\mathbf{x}$  (stationarity condition)

$$\frac{\partial L}{\partial x_1} = 2x_1 + \lambda = 0 \rightarrow \lambda = -2x_1$$

$$\frac{\partial L}{\partial x_2} = 2x_2 + \lambda = 0 \rightarrow \lambda = -2x_2$$

$$x_1 = x_2$$

# Inequality constrained optimization: *Example 1*

3. Use the complementary slackness

**Case 1:**  $\lambda > 0$  and  $g(x)$   
 $= 0$

$$x_1 + x_2 - 1 = 0 \rightarrow 2x_1 = 1$$

$$x_1 = x_2 = \frac{1}{2}$$

4a. Check if the dual feasibility condition is satisfied

$$\lambda = -2x_1 = -1$$

This solution **does NOT satisfy the dual feasibility condition** and it is thus **unfeasible**

# Inequality constrained optimization: *Example 1*

3. Use the complementary slackness

**Case 2:**  $\lambda = 0$  and  $g(x) < 0$

$$\lambda = -2x_1$$

$$x_1 = x_2 = 0$$

4b. Check if the primal feasibility condition is satisfied:

$$g(x_1, x_2) = 0 + 0 - 1 = -1 < 0$$

The solution is feasible and therefore the optimal!



# Inequality constrained optimization: *Example 1*

Summary of the KKT conditions for the problem:

$$\left. \begin{aligned} \lambda &= -2x_1 \\ \lambda &= -2x_2 \end{aligned} \right\} \text{Stationarity}$$

$$x_1 + x_2 - 1 \leq 0 \quad \text{Primal feasibility}$$

$$\lambda \geq 0 \quad \text{Dual feasibility}$$

$$\lambda(x_1 + x_2 - 1) = 0 \quad \text{Complementary slackness}$$



# Gradient descent

- In machine learning, some of the optimization problems suffer from the following challenges:
  - A closed form solution may be computationally intractable
    - Functions are complex: no easy derivative = 0 solution.
    - High dimensions: solving  $\nabla f(x)=0$  analytically is impossible.
  - All the data may not be available at the time of training (e.g. streaming data)
- Gradient descent is an iterative minimization technique for differentiable functions on a domain
- **Intuition:** the function  $f(\mathbf{x})$  decreases the fastest by going from  $x_n$  to  $x_{n+1}$  in the opposite direction of the gradient of  $f(\mathbf{x})$

$$\mathbf{x}_{n+1} = \mathbf{x}_n - \alpha_n \nabla f(\mathbf{x}_n)$$

# Learning rate

- The choice of the learning rate  $\alpha_n$  plays an important role in the success of the minimization:
  - If  $\alpha_n$  is **too small** then the number of iterations necessary to reach the minimum will be too large
  - If  $\alpha_n$  is **too large** then we may overshoot the minimum
- It can be set to a constant or varied iteratively to improve the performance of the algorithm

# Momentum

- The gradient descent technique is very susceptible to local minima, therefore, it can be modified by adding a momentum term

$$\mathbf{x}_{n+1} = \mathbf{x}_n - \alpha_n g(\mathbf{x}_n)$$

Where:

$$g(\mathbf{x}_n) = \mu g(\mathbf{x}_{n-1}) + \nabla f(\mathbf{x}_n)$$

$\mu$  = momentum constant

- The momentum constant also needs to be fine tuned along with the learning rate to each problem, typically picked as a value between 0 and 1